



Methanobrevibacter boviskoreani JH1^T growth on alcohols allows development of a high throughput bioassay to detect methanogen inhibition

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ABSTRACT

Rumen methanogenic archaea use by-products of fermentation to carry out methanogenesis for energy generation. A key fermentation by-product is hydrogen (H₂), which acts as the source of reducing potential for methane (CH₄) formation in hydrogenotrophic methanogens. The *in vitro* cultivation of hydrogenotrophic rumen methanogens requires pressurised H₂ which limits the ability to conduct high-throughput screening experiments with these organisms. The genome of the hydrogenotrophic methanogen *Methanobrevibacter boviskoreani* JH1^T harbors genes encoding an NADP-dependent alcohol dehydrogenase and a F₄₂₀-dependent NADP reductase, which may facilitate the transfer of reducing potential from ethanol to F₄₂₀ via NADP. The aim of this study was to explore the anaerobic culturing of JH1^T without pressurised H₂, using a variety of short chain alcohols. The results demonstrate that in the absence of H₂, JH1^T can use ethanol, 1-propanol, and 1-butanol but not methanol, as a source of reducing potential for methanogenesis. The ability to use ethanol to drive CH₄ formation in JH1^T makes it possible to develop a high throughput culture-based bioassay enabling screening of potential anti-methanogen compounds. The development of this resource will help researchers globally to accelerate the search for methane mitigation technologies for ruminant animals. Global emissions pathways that are consistent with the temperature goal of the Paris Agreement, rely on substantial reductions of agricultural greenhouse gasses, particularly from ruminant animals.

Introduction

Methane emissions from ruminant livestock are the single largest source of agricultural greenhouse gas emissions globally and the development of methanogen inhibiting technologies that do not impact animal productivity are an active area of research (Mizrahi et al., 2021). In the rumen, hydrogenotrophic methanogens use H₂ or formate as a source of reducing potential to reduce carbon dioxide (CO₂) to CH₄, thereby generating membrane potential for ATP biosynthesis (Attwood et al., 2020). Hydrogenotrophic *Methanobrevibacter* species are the dominant rumen methanogens (Henderson et al., 2015) and their H₂ consumption ensures that fermentation remains thermodynamically favourable (Janssen 2010; Ungerfeld 2020).

Methanogens can be difficult to culture and rumen *Methanobrevibacter* spp are strictly anaerobic and dependent on H₂ for growth. Consequently, culture vessels are pumped with H₂:CO₂ (4:1) and

pressurised to 180 kPa to simulate the level of dissolved H₂ within the rumen (Leahy et al., 2010). This makes it challenging to develop large scale or high-throughput culture-based assays that can be used to screen for anti-methanogen activity. Such assays are essential to test the ability of chemical compounds or microbial metabolites such as bacteriocins for the ability to inhibit methanogen growth, or to help select strains suitable for development as direct-fed microbial cultures that could limit CH₄ emissions (Doyle et al., 2019). Methanogenic archaea that can grow on CO₂ and ethanol or isopropanol as sole energy sources have been described (Berk and Thauer, 1997) and it has been shown that H₂ and formate may not be the only sources of reducing potential for some hydrogenotrophic *Methanobrevibacter* species (Leahy et al., 2010; Weimar et al., 2017). Analysis of the genome sequence of the rumen methanogen *Methanobrevibacter* sp. AbM4 identified genes for alcohol and aldehyde dehydrogenases (Leahy et al., 2013), which suggested that AbM4 may be able to use alcohols, such as methanol and ethanol, as an

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alternative source of reducing potential for methanogenesis. The supplementation of both ethanol and methanol at 20 mM allowed AbM4 to grow without pressurised H₂ (Weimar et al., 2017). The presence of homologues of these genes in other methanogens has been shown to allow ethanol to be used as a source of reducing power for CH₄ production (Hoedt et al., 2016; Kelly et al., 2019).

The AbM4 culture is not available from a public culture collection and among the sequenced rumen methanogen genomes, *Methanobrevibacter boviskoreani* JH1^T (Lee et al., 2013b) has the highest degree of similarity to AbM4, with a 16S rRNA identity of 99.8%, indicating that they belong to the same species. *M. boviskoreani* JH1^T was previously reported to use H₂/CO₂ and formate/CO₂ for growth, but not ethanol or methanol (Lee et al., 2013a). However, the NADP-dependent alcohol dehydrogenase, aldehyde dehydrogenase and F₄₂₀-dependent NADP reductase genes are present in the *M. boviskoreani* JH1^T genome and suggest it may have the ability to utilize alcohols for reducing potential in the absence of H₂. The aim of this study was to explore and optimize the growth of *M. boviskoreani* JH1^T without H₂ to support the development of a high throughput bioassay enabling screening of this methanogen against compounds or microbial cultures with potential anti-methanogen activity.

Materials and methods

Methanogen growth

Methanobrevibacter boviskoreani JH1^T (DSM 25,824, Lee et al., 2013a) was obtained from DSMZ, (Braunschweig, Germany) and grown in BY basal liquid medium with the following composition (per L): Salt solution A 170 mL, Salt solution 2B 170 mL, Centrifuged rumen fluid 300 mL, Distilled H₂O 360 mL, Resazurin (0.1% w/v) 10 drops, Yeast extract 1.0 g. Salt solution A contains (per 2.5 L); NaCl 15.0 g, KH₂PO₄ 7.5 g, (NH₄)₂SO₄ 3.75 g, CaCl₂·2H₂O 1.98 g, MgSO₄·7H₂O 3.0 g while Salt solution 2B contains (per 2.5 L) K₂HPO₄ 15.0 g. The BY basal medium was supplemented with trace element solution SL10 (1 mL/L, Widdel et al., 1983) and selenite/tungstate solution (1 mL/L, Tschech and Pfennig 1984). The medium was boiled and gassed with O₂-free 100% CO₂ and NaHCO₃ (5 g/L), and L-cysteine-HCl (0.5 g/L) were added after it cooled. BY medium (4.5 mL) was dispensed into 15 mL Hungate tubes while gassed with O₂-free 100% CO₂ and sealed with butyl rubber bungs, before being sterilised by autoclaving for 20 min at 121°C. Additional supplements added after autoclaving included: coenzyme M (10 µM final conc.), sodium acetate (20 mM final conc.), sodium formate (60 mM final conc.), ethanol (20, 50, 100, 200 or 400 mM final conc.), methanol (200 mM final conc.), 1-propanol (200 mM final conc.), isopropanol (200 mM final conc.), 1-butanol (200 mM final conc.), sodium DL-lactate (200 mM final conc.), and 0.1% (v/v) of vitamin 10 concentrate (per L; 4 aminobenzoate 40 mg, D (+) biotin 10 mg, nicotinic acid 100 mg, hemicalcium D (+)-pantothenate 50 mg, pyridoxamine hydrochloride 150 mg, thiamine chloride hydrochloride 100 mg, cyanocobalamin 50 mg, D, L-6,8-thioctic acid 30 mg, riboflavin 30 mg, folic acid 10 mg). For the microtitre plate assay a vitamin 10/coenzyme M solution was made by combining 1 mL of the Vitamin 10 stock solution with 1 mL of a 10 mM coenzyme M stock solution and diluting to 10 mL with sterile growth medium. The supplements were gassed with O₂-free 100% N₂ gas for 15 min and sterilised by passage through a 0.2 µm pore size sterile filter. Tubes containing BY medium were inoculated with 10% v/v of actively growing culture, the head space was then gassed with either O₂-free H₂:CO₂ (4:1) or CO₂ to 180 kPa.

Assessment of growth

The culture OD (optical density) of *M. boviskoreani* JH1^T was measured at 600 nm by inserting the tubes directly into a Spectronic 200 spectrophotometer (Thermo Fisher Scientific, MA, USA). The growth

was also monitored by CH₄ production and H₂ consumption. Samples of the head-space gas (0.2 mL) were taken by syringe and measured by gas chromatography using an aerograph 660 (Varian Associates, CA, USA) fitted with a Porapak Q 80/100 mesh column (Waters Corporation, MA, USA) and a thermal conductivity detector operated at 100°C. Alcohol and volatile fatty acid concentrations in the medium were determined with a nitroterephthalic acid-modified polyethylene glycol column (Zebron ZB-WAX-Plus Capillary GC Column; 30 m by 0.53 mm by 0.25 µm film thickness; Phenomenex, CA, USA) attached to a Hewlett-Packard 6890 gas chromatography system. Gas chromatography was carried out as previously described (Attwood et al., 1998).

Microtitre plate assay of *M. boviskoreani* JH1^T

The *M. boviskoreani* JH1^T inoculum for seeding the plate assay was grown in Hungate tubes containing 9 mL BY medium supplemented with sodium formate (60 mM final conc.), ethanol (200 mM final conc.), coenzyme M and vitamin 10 concentrate. The tubes were pressurized with O₂-free CO₂ to 180 kPa and incubated at 39 °C without shaking until visible turbidity appeared (3 to 5 days). The freshly grown cultures were checked using wet mounts under fluorescence microscopy, and *Mbb. boviskoreani* JH1^T appears as short ovoid-shaped rods that fluoresce green under ultraviolet illumination. Culture contamination was checked by inoculating a sample of the culture into 9 mL BY medium supplemented with 5 mM glucose and incubating at 39 °C for one day. If no turbidity was seen, then the culture was considered uncontaminated.

For development of the bioassay all of the assay components, except the JH1^T inoculum, were added via CO₂-flushed syringes and needles to 3.75 mL BY plus formate (60 mM final conc.) medium in sterile 7.5 mL Hungate tubes in the proportions indicated in Table 1. Each tube was inoculated with freshly grown JH1^T culture, incubated for 1 hr at 39 °C, then moved inside an anaerobic chamber (98% CO₂- 2% H₂ atmosphere; Coy Laboratory Products, USA) and dispensed into wells of a Falcon 96 well polystyrene plate (Corning, USA). The inoculated 96 well plate was placed into an AnaeroPack 2.5 L Rectangular Jar (Thermo Fisher Scientific) with an Anaeropack-Anaero Anaerobic Gas Generator (Thermo Fisher Scientific), the lid sealed, and the jar removed from the anaerobic chamber for incubation at 39 °C. The plate was observed daily through the transparent jar, and when the *M. boviskoreani* JH1^T control showed visible turbidity, the plate was removed from the jar and the optical density (OD₆₀₀) recorded after 5 s shaking in a Multiskan FC Microplate Photometer (Thermo Fisher Scientific). The absorbance readings of the media control wells were subtracted as background, and the % inhibition of *M. boviskoreani* JH1^T growth relative to the JH1^T control wells calculated.

Nisin preparation

Anaerobic double distilled water (24.9 mL) was mixed with 0.1 mL of 1 M HCl and 187.5 mg NaCl (0.004 M HCl and 0.75% NaCl) and used to dissolve 1 g of 2.5% nisin powder (Sigma-Aldrich) while bubbling with O₂-free 100% CO₂. The nisin preparation was sterilised through a 0.22 µm filter into an autoclaved anaerobic (100% CO₂-flushed) serum bottle and stored at 4 °C.

Table 1
Microtitre plate assay components.

Component	Media control	JH1 ^T control	Nisin
BY medium plus formate	150 µL	150 µL	150 µL
Phosphate buffer (1 M)	90 µL	80 µL	70 µL
EtOH (10 M)	5 µL	5 µL	5 µL
Vitamin /CoM solution	5 µL	5 µL	5 µL
Nisin (1 mg/mL, 300 µM)	0	0	10 µL
JH1 ^T inoculum	0	10 µL	10 µL
Total volume	250 µL	250 µL	250 µL

Genome analysis

Comparative genome analysis and determination of the pairwise average nucleotide identity of the *Methanobrevibacter* strains AbM4 (Leahy et al., 2013) and JH1^T (Lee et al., 2013b) was performed using the online resources of the United States Department of Energy, Joint Genome Institute Integrated Microbial Genomes and Microbiomes resource (Chen et al., 2021). A mummerplot was generated using the program nucmer from the software package MUMmer (Version 3.23, Kurtz et al., 2004) with the genome sequence of JH1^T (GenBank assembly accession: GCA_000320505.2) and Abm4 (GenBank assembly accession: GCA_000404165.1).

Results

Comparison between *M. boviskoreani* JH1^T and AbM4

We have previously documented the high similarity between the finished genome of *M. sp.* AbM4 (1998,189 bp, GenBank ID CP004050) and the 54 contig draft genome assembly of *M. boviskoreani* JH1^T (2045,801 bp, GenBank ID BAGX000000000) (Leahy et al., 2013). Both strains have a G + C content of 29%. The pairwise average nucleotide identity between *M. sp.* AbM4 and *M. boviskoreani* JH1^T is 98.2%, and the genomic content of *M. boviskoreani* JH1^T is almost identical to *M. sp.*

AbM4 (Fig. 1), with the biggest difference being the presence of a prophage in the *M. boviskoreani* JH1^T genome. AbM4 has been shown to grow in the presence of 20 mM methanol and ethanol without pressurised H₂:CO₂ (Weimar et al., 2017). *M. boviskoreani* JH1^T possesses homologues of the NADP-dependent F₄₂₀ dehydrogenase (NoneDRAFT_00,991), alcohol dehydrogenase (NoneDRAFT_00,994), and aldehyde dehydrogenase genes (NoneDRAFT_00993) whose presence enables alcohol-fueled growth by strain AbM4 (Hoedt et al., 2016). The ability to utilize alcohols depends on the two dehydrogenases with the alcohol dehydrogenase catalysing the conversion of ethanol to acetaldehyde, and the neighboring aldehyde dehydrogenase converting acetaldehyde to acetate. Consequently, *M. boviskoreani* JH1^T was predicted to be able to utilize alcohols as a source of reducing potential with NADP and F₄₂₀ as key cofactors.

Growth and methane production of JH1^T with ethanol

The potential ability of *M. boviskoreani* JH1^T to use ethanol as a source of reducing potential for growth and CH₄ production was studied under nine conditions (Fig. 2), using combinations of formate, ethanol, and methanol with or without H₂. In the presence of H₂, cultures supplemented with ethanol or methanol reached comparable OD₆₀₀ to the control (Fig. 2A) and consumed similar levels of H₂ (Fig. 2C). Ethanol supplementation allowed *M. boviskoreani* JH1^T to grow without H₂ and

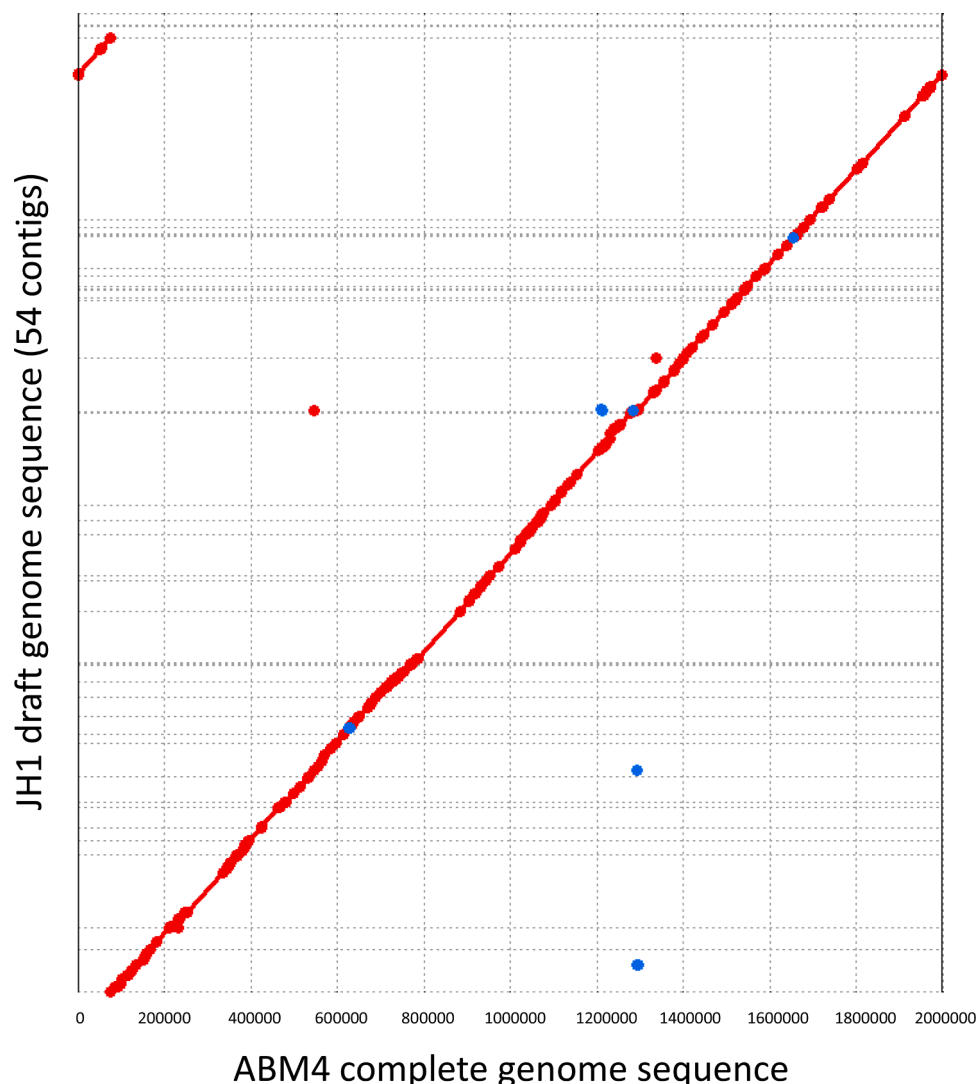


Fig. 1. MUMmerplot comparing the genome assemblies of *Methanobrevibacter* strains AbM4 and *Methanobrevibacter boviskoreani* JH1^T.

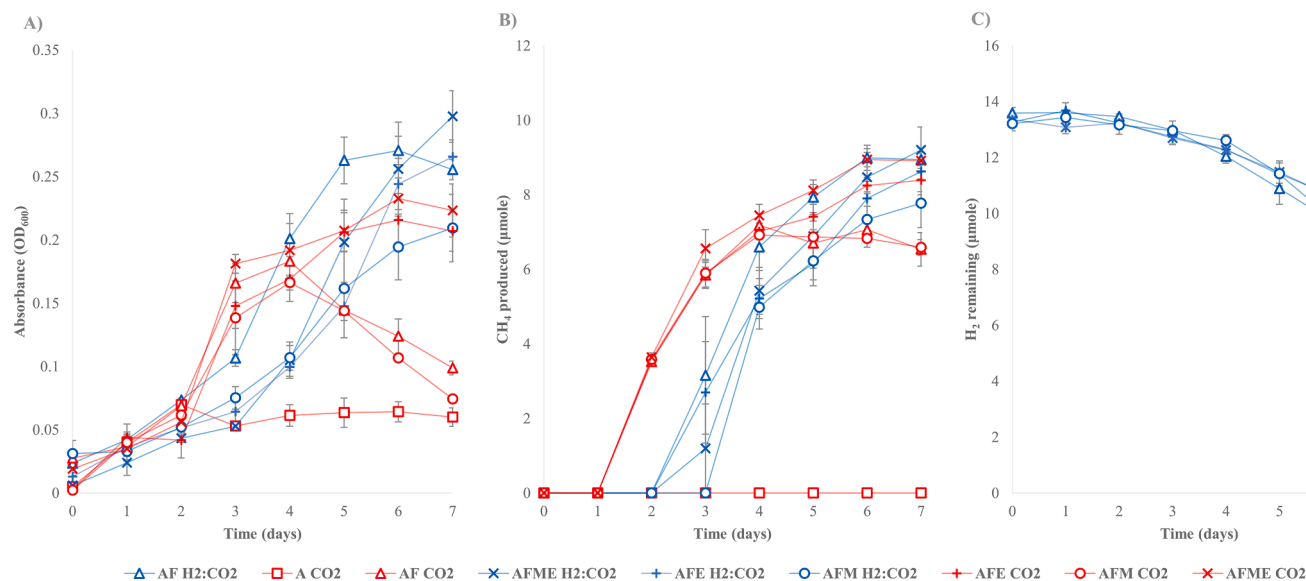


Fig. 2. Growth of *M. boviskoreani* JH1^T with different combinations of formate (F, 60 mM), methanol (M, 20 mM), ethanol (E, 20 mM) and either CO₂ (180 kPa, red) or H₂:CO₂ (4:1; 180 kPa, blue) in BY basal medium. All replicates contained acetate (A, 20 mM). A) Optical density of *M. boviskoreani* JH1^T. B) Amount of CH₄ produced per mL of headspace by *M. boviskoreani* JH1^T over seven days. C) H₂ per mL of headspace remaining from *M. boviskoreani* JH1^T growth. Error bars represents ±SEMs (standard error of the means).

produce similar levels of CH₄ to the cultures grown with H₂ (Fig. 2B). Supplementation with formate alone allowed *M. boviskoreani* JH1^T to produce CH₄ without H₂, but once the formate was consumed on day four the OD₆₀₀ declined (Fig. 2A). A similar result was observed with supplementation of formate and methanol together without H₂ indicating that methanol alone did not support growth.

The optimal concentration of ethanol supplementation was explored as shown in Fig. 3. *M. boviskoreani* JH1^T was subcultured twice without formate to remove any residual formate, and subsequently cultivated in the absence of H₂ with ethanol concentrations of 20, 50, 100, 200, and 400 mM. The growth of *M. boviskoreani* JH1^T supplemented with 400 mM of ethanol gave the highest stationary phase OD₆₀₀ (Fig. 3A) and CH₄ production per mL of headspace (Fig. 3B), but supplementation with 200 mM ethanol also gave good growth. The results from GC analysis indicated that at 200 mM ethanol, all the ethanol was converted into acetate, while at 400 mM ethanol approximately 300 mM of the ethanol was converted to acetate (Table 2). The high amount of acetate

produced lowered the culture pH to 4.27 ± 0.2 , which may explain why a decline in OD₆₀₀ was observed after day 4 (Fig. 3A). Therefore, supplementation with 200 mM ethanol alone was sufficient for *M. boviskoreani* JH1^T to grow and produce CH₄ in the absence of formate or acetate or H₂ (Fig. 3A and 3B), and was the level used in further studies.

Growth and methane production of JH1^T with other alcohols and lactate

To determine if *M. boviskoreani* JH1^T was specialized for ethanol consumption, other short chain alcohols (methanol, 1-propanol, isopropanol, 1-butanol) and lactate were supplemented at 200 mM (Fig. 4). Results showed that *M. boviskoreani* JH1^T grew equally well on ethanol and 1-propanol, while growth was slower on 1-butanol. GC analysis indicated that 1-propanol was completely converted to propionic acid, and 1-butanol was completely converted to butyric acid (Table 2). Based on the predicted pathway of ethanol metabolism, 1-propanol and 1-

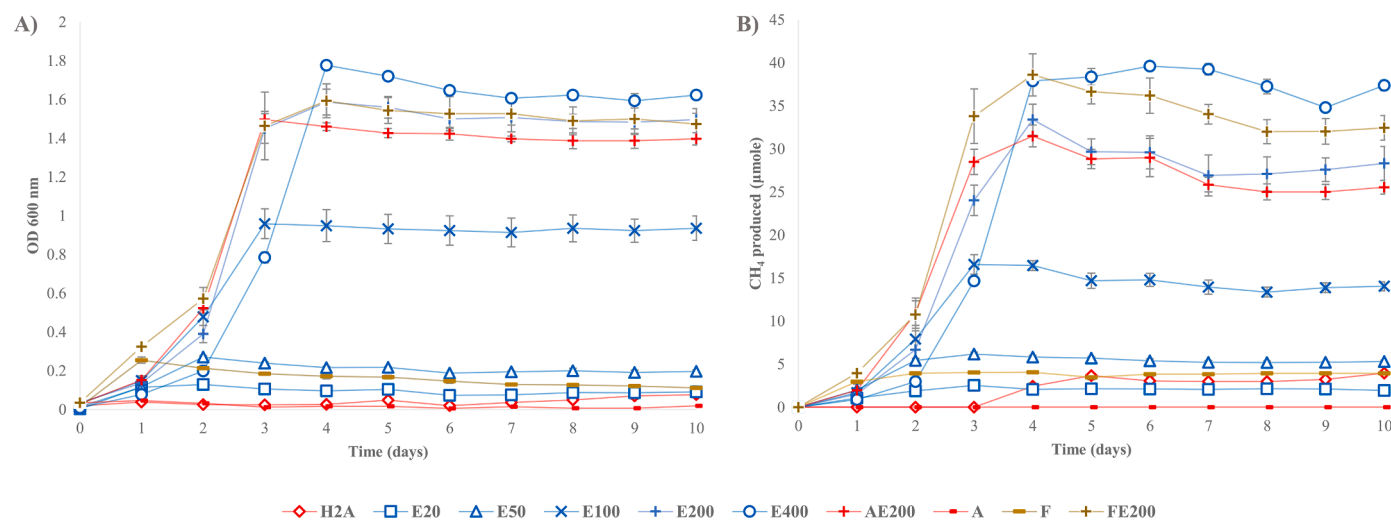


Fig. 3. Growth of *M. boviskoreani* JH1^T with 20 mM, 50 mM, 100 mM, 200 mM, or 400 mM of ethanol (E, blue), formate (60 mM, F/FE200, brown), and acetate (20 mM, A/AE200, red) with CO₂ (180 kPa) in BY basal medium. A) Optical density of *M. boviskoreani* JH1^T. B) Amount of CH₄ produced per mL of headspace by *M. boviskoreani* JH1^T over ten days. Error bars represents ±SEMs.

Table 2

Average concentration of alcohols, VFAs and pH after culture growth.

	Acetic acid (mM)	Methanol (mM)	Ethanol (mM)	1-Propanol (mM)	Propionic acid (mM)	1-Butanol (mM)	Butyric acid (mM)	pH
H ₂	55.6	0	0	0	0	0	0	6.61
E20	53.8	0	0	0	0	0	0	6.46
E50	70.0	0	0	0	0	0	0	6.21
E100	154.8	0	0	0	0	0	0	4.93
E200	246.9	0	0	0	0	0	0	4.49
E400	322.2	0	106.4	0	0	0	0	4.27
EA	257.9	0	0	0	0	0	0	4.60
A	61.6	0	0	0	0	0	0	6.58
F	28.2	0	0	0	0	0	0	6.96
EF	233.9	0	0	0	0	0	0	4.99
M 200	28.5	203.8	0	0	0	0	0	6.59
ME200	216.9	186.0	0	0	0	0	0	4.56
PrOH200	25.7	0	0	0	189.7	0	0	4.67
P2OH200	29.5	0	0	0	0	0	0	6.57
B200	26.6	0	0	0	0	0	190.5	4.69
L200	28.6	0	0	0	0	0	0	6.56

M. boviskoreani JH1^T was grown with hydrogen (H₂), 20 mM ethanol (E20), 50 mM ethanol (E50), 100 mM ethanol (E100), 200 mM ethanol (E200), 400 mM ethanol (E400), 200 mM ethanol/20 mM acetate (EA), 20 mM acetate (A), 60 mM formate (F), 200 mM ethanol/60 mM formate (EF), 200 mM methanol (M200), 200 mM methanol/200 mM ethanol (ME200), 200 mM 1-propanol (PrOH200), 200 mM isopropanol (P2OH200), 200 mM 1-butanol (B200) and 200 mM lactate (L200) on CO₂ (180 kPa) in BY basal medium.

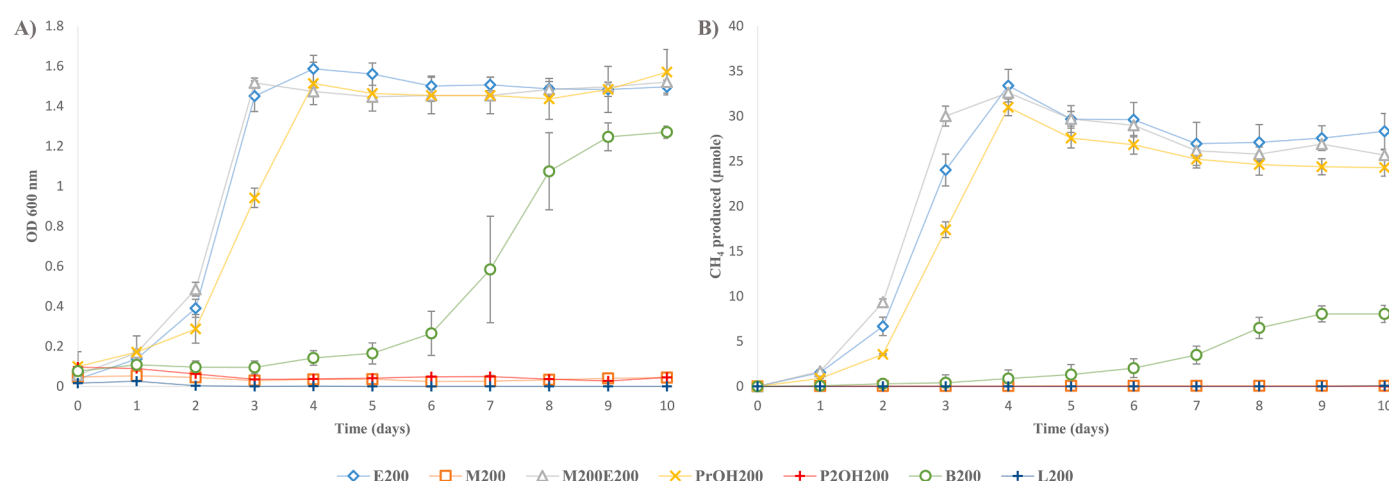


Fig. 4. Growth of *M. boviskoreani* JH1^T with 200 mM ethanol (E200), 200 mM methanol (M200), 200 mM 1-propanol (PrOH200), 200 mM isopropanol (P2OH200), 200 mM 1-butanol (B200) and 200 mM lactate (L200) with CO₂ (180 kPa) in BY basal medium. A) Optical density of *M. boviskoreani* JH1^T. B) Amount of CH₄ produced per mL of headspace by *M. boviskoreani* JH1^T over ten days. Error bars represents ± SEMs.

butanol may have been converted to propionaldehyde and butyraldehyde respectively as intermediates. *M. boviskoreani* JH1^T was unable to utilize isopropanol, methanol, or lactate (Fig. 4), and co-supplementation of methanol and ethanol did not result in any observed synergy.

The requirement of CO₂ for methanogenesis was verified by growing *M. boviskoreani* JH1^T supplemented by 200 mM ethanol alone, with either CO₂ or N₂ in the headspace. *M. boviskoreani* JH1^T failed to grow with only N₂ in the headspace (Fig. 5), indicating that methanogenesis in the absence of H₂ still requires CO₂.

Development of a microtitre plate bioassay to screen methanogen inhibitors

We used the ability of *M. boviskoreani* JH1^T to grow and produce CH₄ using ethanol plus CO₂ to develop a 96 well microtitre plate growth inhibition assay suitable for the high throughput screening of microbial cultures or metabolites. To test the utility of this bioassay for detecting methanogen inhibition we used nisin, a bacteriocin produced by *Lactococcus lactis* and commercially available in a semi-pure form. Nisin has been shown to inhibit CH₄ production in rumen *in vitro* and *in vivo* assays (Doyle et al., 2019) but it is not known from these studies if nisin affects the methanogens or other rumen microbes that produce methanogenesis

substrates. The set-up of the microtitre plate assay is shown in Table 1. For the inhibition assay the absorbance readings of the media control wells were subtracted as background, and the % inhibition of *M. boviskoreani* JH1^T growth when exposed to nisin relative to growth of JH1^T in the control wells was calculated. Using this bioassay, we were able to demonstrate that exposure to nisin resulted in almost complete inhibition of *M. boviskoreani* JH1^T growth relative to the control (Fig. 6).

Discussion

Methane resulting from enteric fermentation in ruminant livestock is a major source of agricultural greenhouse gasses. Consequently, the need to control CH₄ emissions has become a priority resulting in increased urgency to develop strategies that can practically mitigate CH₄ from ruminant animals (Reisinger et al., 2021). Several different approaches have been used to specifically target methanogens in the rumen including the use of feed additives, certain seaweeds, and development of anti-methanogen vaccines (Beauchemin et al., 2020). The use of microbial cultures (particularly lactic acid bacteria) or their metabolites has also been proposed (Doyle et al., 2019). However, methanogens are difficult to culture, and a major limitation is the inability to perform high throughput screening of cultures or their

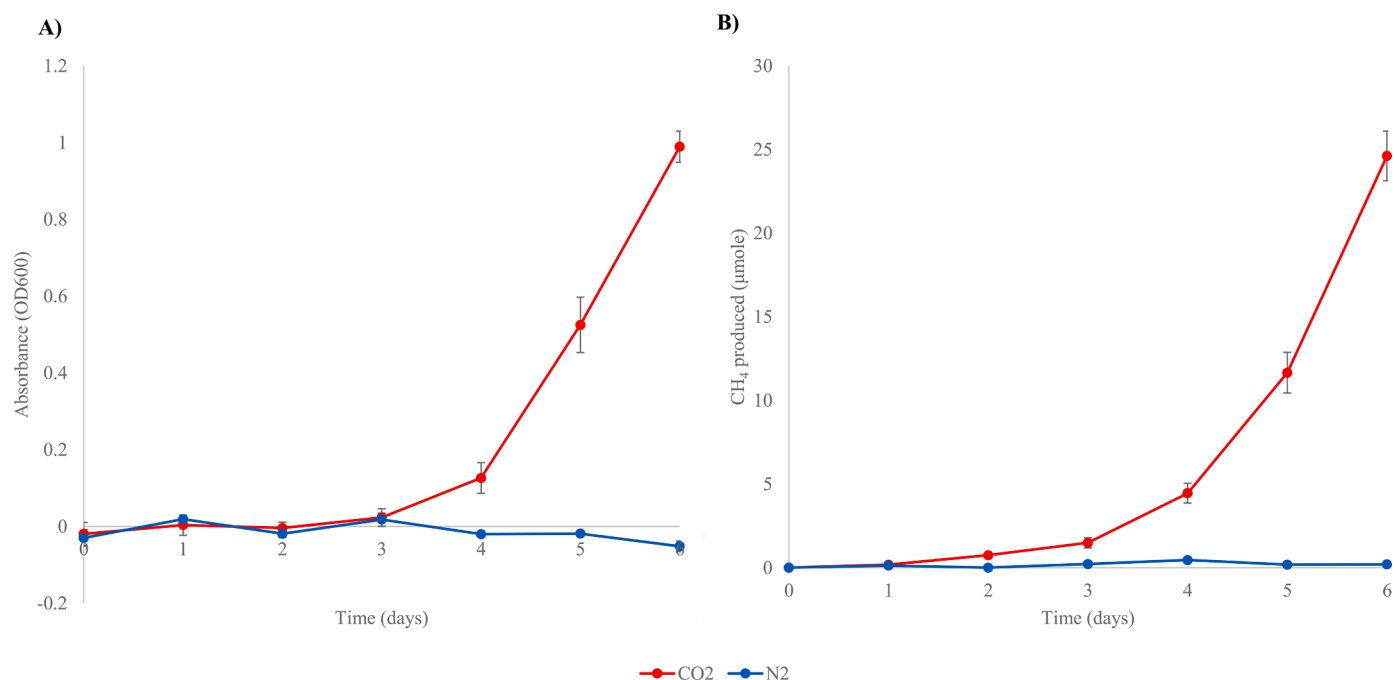


Fig. 5. Growth of *M. boviskoreani* JH1^T with 200 mM ethanol with CO₂ (180 kPa, red) or N₂ (180 kPa, blue) in BY basal medium. A) Optical density of *M. boviskoreani* JH1^T. B) Amount of CH₄ produced per mL of headspace by *M. boviskoreani* JH1^T over six days. Error bars represents ±SEMs.

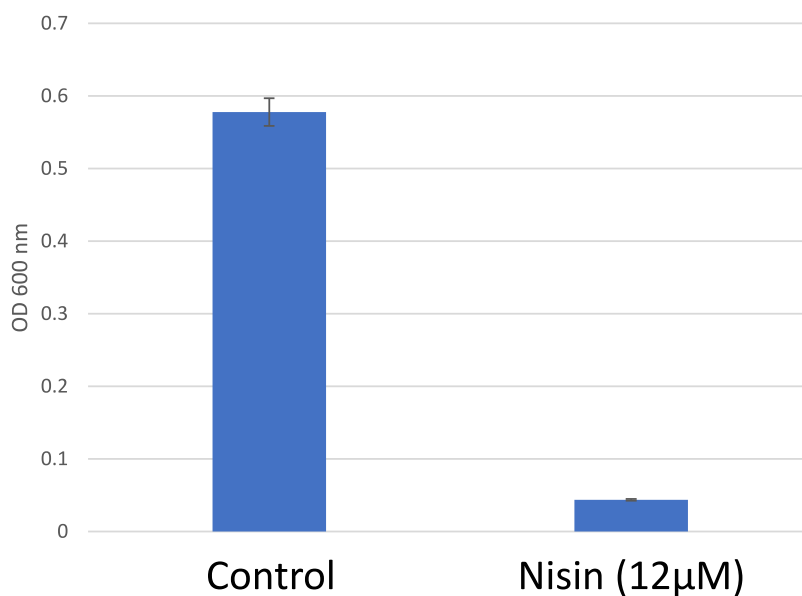


Fig. 6. Effect of nisin on the growth of *M. boviskoreani* JH1^T in microtitre plates. Cells were grown in BY basal medium supplemented with 200 mM ethanol and 60 mM formate with and without nisin (Table 1). Error bars represents ±SEMs.

metabolites to select those capable of methanogen inhibition. Therefore, new tools are needed that can contribute to the acceleration of CH₄ mitigation research (Leahy et al., 2022).

Here we have shown that *M. boviskoreani* JH1^T can grow without H₂ in the presence of ethanol, 1-propanol, or 1-butanol, and that supplementation with ethanol was able to greatly enhance the growth of *M. boviskoreani* JH1^T in the absence of pressurised H₂. Ethanol is a by-product of rumen fermentation, with the normal physiological concentration of ruminal fluid ethanol being 556 ± 450 μM according to the Bovine Metabolome Database (Foroutan et al., 2020). Therefore, it was surprising to find *M. boviskoreani* JH1^T could grow at ethanol concentration more than 200-fold above the physiological concentration,

suggesting the *M. boviskoreani* JH1^T may have evolved as a specialized ethanol consuming methanogen or is able to partner with ethanol producing bacteria (Ungerfeld, 2020).

The ability of *M. boviskoreani* JH1^T to grow and produce CH₄ in the absence of pressurised H₂ enabled the development of a microtitre plate-based methanogen inhibition bioassay. We tested this bioassay using the bacteriocin nisin and showed that purified nisin was able to successfully inhibit the growth of *M. boviskoreani* JH1^T. Although nisin has previously been shown to inhibit a non-rumen methanogen (*Methanobacterium*) using an agar diffusion assay (Hammes et al., 1979) this is the first time that such a direct effect of nisin on a rumen methanogen has been demonstrated. We plan to use this bioassay to screen

metabolites from rumen microbes and lactic acid bacteria (Doyle et al., 2019) for the ability to inhibit methanogens. Consequently, we believe that this *Methanobrevibacter* culture will be a useful tool to develop high-throughput screening approaches targeted towards inhibition of rumen methanogenesis.

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Authors' contribution

All authors checked and approved final version of the manuscript.

CRediT authorship contribution statement

Yang Li: Investigation, Data curation, Writing – original draft. **Laureen Crouzet:** Investigation. **William J. Kelly:** Conceptualization, Funding acquisition, Methodology, Supervision, Writing – original draft, Writing – review & editing. **Peter Reid:** Investigation. **Sinead C. Leahy:** Conceptualization, Funding acquisition, Methodology, Supervision, Writing – original draft, Writing – review & editing. **Graeme T. Attwood:** Conceptualization, Funding acquisition, Methodology, Supervision, Writing – original draft, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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